

Predictive Model Plan

1. Model Logic (Generated with GenAI)

The predictive model selected for this task is Logistic Regression.

This model is designed to predict whether a customer is likely to become delinquent based on financial and behavioral information available in the dataset.

Logistic regression is appropriate because delinquency prediction is a binary classification problem where the outcome is either delinquent (1) or non-delinquent (0).

The model workflow begins by collecting customer data such as :Income, Credit_Score, Credit_Utilization, Missed_Payments, Debt_to_Income_Ratio, Loan_Balance, Employment_Status, Account_Tenure, and payment history from Month_1 to Month_6.

The data is then preprocessed by handling missing values, standardizing inconsistent entries, and converting categorical variables such as Employment_Status, Credit_Card_Type, and Location into machine-readable formats.

After preprocessing, the logistic regression model analyzes the relationship between the input variables and the target variable Delinquent_Account.

The model calculates a probability score between 0 and 1 that represents the likelihood of a customer becoming delinquent.

Customers with higher probability scores are classified as high-risk customers and can be prioritized for proactive intervention strategies by Geldium’s Collections team.

The top input features used in the model include:

Input Feature	Reason for Importance
Missed_Payments	Customers with more missed payments are more likely to become delinquent
Credit_Utilization	High credit utilization may indicate financial stress
Debt_to_Income_Ratio	Higher debt burden can reduce repayment ability
Credit_Score	Lower credit scores may reflect poor credit reliability
Month_1 to Month_6 Payment History	Recent payment behavior helps identify repayment trends

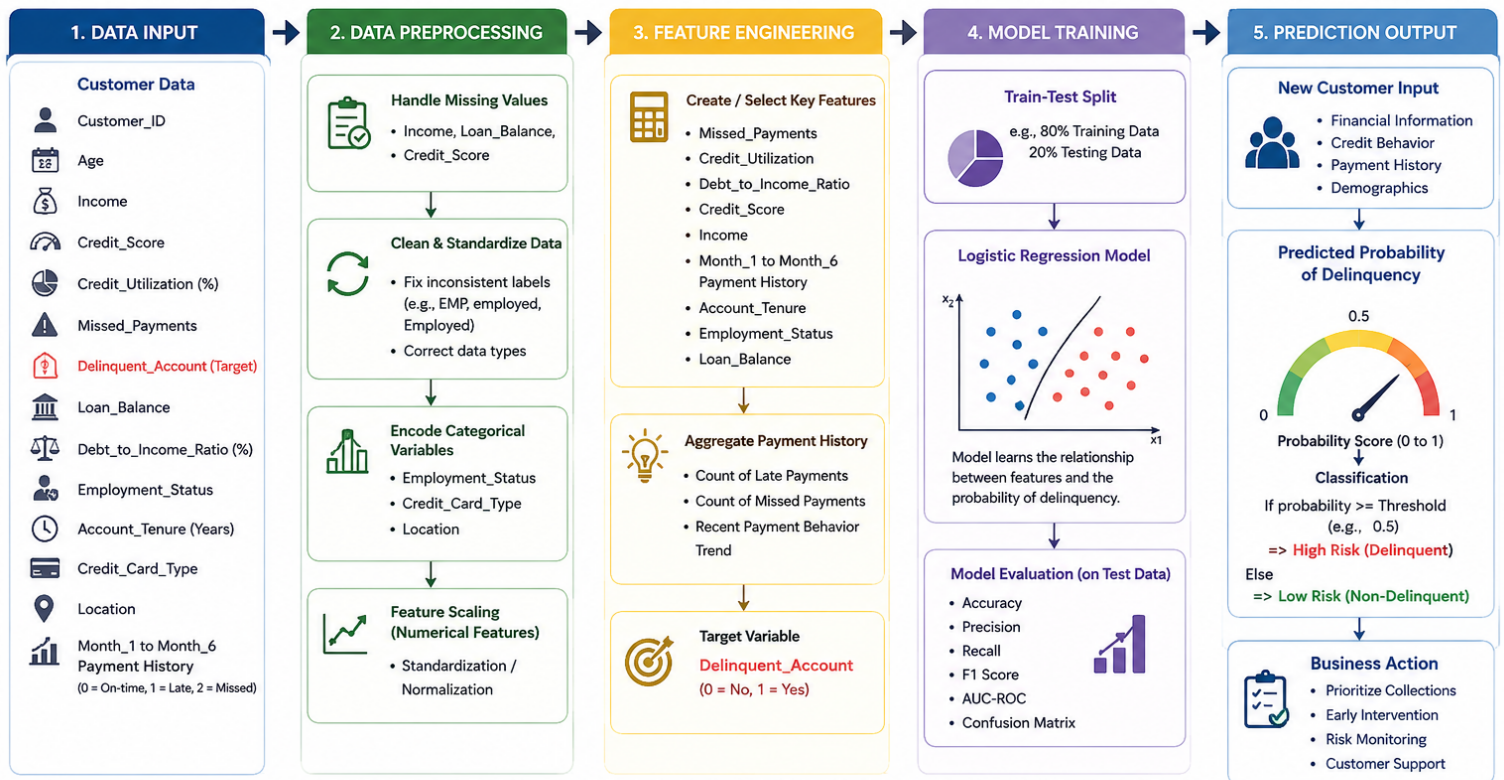
The dataset also revealed several data quality concerns that should be addressed before model deployment.

Missing values were observed in fields such as Income, Loan_Balance, and Credit_Score.

Inconsistent employment labels such as “EMP,” “Employed,” and “employed” were also identified and would require standardization. Additionally, some Credit_Utilization values appeared unusually high, which may indicate outliers or data inconsistencies.

Predictive Model Logic

Logistic Regression Model for Predicting Customer Delinquency



WORKFLOW OVERVIEW

The model takes customer financial and behavioral data, preprocesses and transforms it, selects important features, trains a logistic regression model, and outputs the probability of delinquency to support risk-based decision making.

Fig.- Figure 1: AI-assisted predictive modeling pipeline for customer delinquency risk assessment.

2. Justification for Model Choice

Logistic regression was selected because it is highly suitable for binary classification problems such as predicting whether a customer will become delinquent or non-delinquent. The model is simple, transparent, and easy to interpret, allowing Geldium's Collections and Risk teams to clearly understand how variables such as missed payments, credit utilization, debt-to-income ratio, and payment history contribute to delinquency risk. In financial services, explainability is extremely important because organizations must justify risk assessments and ensure compliance with responsible AI practices.

Key Reasons for Choosing Logistic Regression

- *Suitable for binary classification-*
The model is effective for predicting two possible outcomes: delinquent (1) or non-delinquent (0).
- *High interpretability and transparency-*
Logistic regression clearly shows how input variables influence delinquency predictions, making the model easier for stakeholders to understand.
- *Supports regulatory compliance-*
Financial institutions require explainable AI systems to justify risk assessments and maintain ethical decision-making standards.
- *Works well with structured financial data-*
The dataset contains structured numerical and categorical variables such as income, credit score, payment history, and debt-to-income ratio, which are well-suited for logistic regression.
- *Efficient and scalable-*
The model is computationally efficient, easier to deploy, and can scale effectively across large customer datasets.

Comparison with Other Models

- *Compared to Neural Networks:*
Neural networks may identify deeper and more complex patterns, but they operate as "black-box" models, making predictions difficult to explain and monitor.
- *Compared to Decision Trees:*
Decision trees are also interpretable, but logistic regression provides more stable probability-based outputs and is commonly used in financial risk prediction.

Overall, logistic regression provides the best balance between:

- *accuracy,*
- *explainability,*
- *fairness,*
- *operational efficiency,*
- *and ease of deployment*

for Geldium's delinquency prediction needs.

3. Evaluation Strategy

The predictive model will be evaluated using multiple performance metrics to ensure that it produces accurate, reliable, and fair predictions. Since the model is designed to identify customers at risk of delinquency, it is important to measure not only overall correctness but also how effectively the model identifies high-risk customers while minimizing incorrect predictions.

Performance Metrics

- *Accuracy*
Accuracy measures the overall percentage of correct predictions made by the model. It helps evaluate how well the model performs across all customer records.
- *Precision*
Precision measures how many customers predicted as delinquent are actually delinquent. High precision is important to reduce unnecessary intervention efforts for customers who are not truly high risk.
- *Recall (Sensitivity)*
Recall measures how many actual delinquent customers are correctly identified by the model. This is one of the most important metrics for Geldium because failing to identify high-risk customers could result in financial losses.
- *F1 Score*
The F1 score balances precision and recall, ensuring the model performs consistently across both metrics. It is useful when both false positives and false negatives are important.
- *AUC-ROC Score*
The AUC-ROC metric evaluates how effectively the model distinguishes between delinquent and non-delinquent customers. A higher AUC score indicates better prediction capability.
- *Confusion Matrix*
A confusion matrix will be used to analyze true positives, true negatives, false positives, and false negatives. This helps identify the types of prediction errors the model is making.

Interpretation of Metrics

- *High accuracy indicates strong overall model performance.*
- *High precision means the model avoids flagging too many low-risk customers as delinquent.*
- *High recall ensures the model successfully identifies most high-risk customers.*
- *A balanced F1 score indicates that the model performs well across both precision and recall.*
- *A high AUC-ROC score suggests the model can effectively separate risky customers from safe customers.*

Bias Detection and Fairness Checks

To ensure fairness and responsible AI usage, the model will be monitored for:

- *demographic bias,*
- *proxy discrimination,*
- *and unequal prediction patterns across customer groups.*

Variables such as Location or Employment_Status will be reviewed carefully to ensure they do not unfairly disadvantage certain customers. Fairness testing and human oversight will be included throughout the evaluation process to support ethical financial decision-making.

Ethical Considerations

Since the model will influence customer risk assessments and intervention strategies, ethical AI practices are essential. The model should:

- *avoid unfair discrimination,*
- *provide explainable predictions,*
- *maintain customer privacy,*
- *and support transparent decision-making.*

Human review and continuous monitoring will also be necessary to ensure the model remains accurate, fair, and compliant with financial regulations over time.